

Text-shaped cleaning with explicit padding

Research and implementation pack
Eight numbered documents · Version 1.0 · 23 September 2026

Required behavior: keep the current cleaning option, add an optional lettering-shaped footprint, show the actual edit boundary, and clean within that boundary without requiring text recognition.

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How to use this pack

Start with **01** for the complete decision register and **03** for the mask algorithm. Use **05–07** as the implementation and acceptance-test handoff. Editable Word and repository-friendly Markdown copies are included in the ZIP.

Status: confirmed user requirements are separated from working recommendations, deferred alternatives and open choices. CTD + COO is an experiment to evaluate, not a demonstrated winner. No application changes or checkpoint benchmarks were performed.

The Korean reference requires Korean validation; published Japanese COO results do not establish that coverage. Source links are included throughout and consolidated in Document 08. Public branch URLs are mutable; pin the implementation checkout and exact model artifacts.

DOCUMENT 01

Complete decision register

Confirmed requirements, working directions, deferred ideas and open choices

1. Status and scope

This is the complete discussion register for the three user requests in this conversation. It is not a claim that the app has been changed. Research date: 23 September 2026. Sources were inspected on public main/master branches; an immutable Manga Cleaner commit was not established.

Status key: CONFIRMED means explicitly requested by the user. WORKING DIRECTION means an assistant recommendation consistent with the request, but not a benchmarked or individually approved choice. DEFERRED preserves an earlier idea without making it a dependency. OPEN requires a product choice, experiment or permission check.

Confirmed requirements

C01 - Clean the text properly. The goal is removal of lettering, not transcription. Regular bubble text and sound effects are both in scope.

C02 - Recover missed regions. The user reports that most SFX and some bubble text are missed. Investigate true detector misses separately from detected-but-skipped regions and inadequate erase masks.

C03 - No OCR requirement. This workflow must not require manga-ocr, text recognition, translation, a transcript, or a successful language read. Existing optional reading features need not be deleted from the app.

C04 - Text-shaped affected area. Add a selection/cleaning area that follows the detected lettering or compact text footprint instead of repainting a large surrounding rectangle.

C05 - Small adjustable padding. Expand that footprint slightly around the text. The user has not selected a numeric default, unit convention or maximum.

C06 - The visible footprint governs the edit. The new highlighted area is the area within which changes may be applied. It must not merely decorate a larger hidden erase rectangle.

C07 - Preserve the old option. The new behavior is additional. Do not replace, silently migrate or remove the current cleaning/selection behavior.

C08 - Use the images as references. The first image communicates the desired compact shape; the second communicates unwanted rectangular overreach. They are illustrative references, not a verified before/after app output or pixel-accurate training labels.

C09 - Deliver itemized documentation. Preserve the full discussion and research, including earlier alternatives, caveats and decisions still open, in separate numbered documents.

Current scope interpretation

The earlier request for class detection for “EVERYTHING” is now an operational goal: find and clean all target lettering safely. A large semantic taxonomy is not required to deliver this feature. Detection and pixel segmentation are still required; “no OCR” does not exclude those image-processing tasks.

2. Working technical directions

The following are proposed implementation decisions. They are not presented as already approved changes or measured improvements.

ID	Working direction
R01	Retain CTD as the current text discovery and segmentation baseline.
R02	Evaluate a COO-trained detector for extra SFX coverage; start with DBNet++ detection-only. CTD + COO is the leading experiment, not a proven winner.
R03	Keep the existing ogkalu text/bubble detector for enclosure and complementary candidates; it is neither OCR nor a detailed SFX classifier.
R04	Add a no-recognition, all-text cleaning policy. Bypass the Japanese gate only for that explicitly selected policy; do not remove legacy gating globally.
R05	Keep detection profile, cleaning-area mode, language policy and outside-bubble permission independent.
R06	Retain credible candidates from either detector. Do not require CTD agreement to admit a COO-only SFX.
R07	Test CTD's unused line output, orphan segmentation candidates and overlapping high-resolution passes for missing bubble text.
R08	Refine lettering at source resolution using CTD evidence and the source image; evaluate upstream-style refinement before adding another large model.
R09	Store an immutable base lettering mask. Derive padding from that base, never from the previously padded/applied mask.
R10	Separate detection bounds, refinement/inpainting context, the model's hole and final allowed write support.
R11	In the new mode, every participating engine must respect one authoritative final mask; do not add invisible isolation or denoise margins afterward.
R12	Show the same backend-generated footprint in the UI, use it for cleaning, and persist its identity/hash. Rectangles remain metadata and optional handles.
R13	Preserve disconnected SFX fragments, punctuation and relevant outlines/shadows. Grouping does not authorize filling the space between fragments.
R14	Provide mask correction and explicit review outcomes. An empty or unsafe mask must not silently become a filled detection rectangle.
R15	Preserve layers, undo/redo, source files, export fidelity, region provenance and old-project behavior.
R16	Prototype research checkpoints separately. Ship an optional native runtime adapter only after output parity, device testing and licensing review.
R17	Evaluate complementary recall, artwork damage and human cleanup effort, not only OCR accuracy or detector leaderboard Hmean.
R18	Implement strict mask geometry first; additional SFX detection must feed that safe path rather than make larger erasures.

3. Earlier ideas retained, not silently promoted

ID	Discussed item and current disposition
D01	Manga-ocr together with COO: technically a different recognition stage, but no recognition is needed for the user's current workflow. Keep old optional support; do not add it to the new dependency chain.
D02	PP-OCRv6 medium detector: optional additional discovery experiment, especially on non-Japanese or missed small text; its recognizer is out of scope.
D03	Other ogkalu checkpoints: an A/B test, not an assumption that a larger or non-quantized export wins. Validate each export's contracts.
D04	MTSv3 / ABCNetv2: research comparators, not default bundled choices. Deployment and licensing need separate checks.
D05	COO TRBA recognition and M4C linking: not needed for cleaning. Geometric grouping can exist without recognizing text.
D06	ogkalu YOLOv8m text segmenter: optional mask experiment; verify whether masks represent text strokes or broader text regions.
D07	SAM-TS / Hi-SAM: newly researched stroke-segmentation comparator, not an approved production dependency.
D08	Detailed semantic-role classifier: defer unless users need different treatment of dialogue, SFX, signage, credits or other roles.
D09	Magi v2/v3, MangaLMM and vision-language reviewers: preserve as earlier exploration; exclude from the immediate cleaning feature.
D10	Custom model training: future option if evaluation shows persistent misses or poor masks; requires usable rights and independently annotated holdouts.
D11	Full-page plus roughly 1024-1536 source-pixel crops with 15-25% overlap: earlier experimental starting range, not an agreed optimal setting.
D12	Text-edge padding around 1-3 source pixels: earlier experimental starting range, not a fixed product decision.
D13	Automatic residual-text checks: may use visual or detector evidence without OCR; uncertain results still need human review.
D14	Heavier inpainting/cloud engines: no new requirement. Preserve existing availability/consent rules; do not make them automatic dependencies.

Earlier taxonomy, preserved for future work

Role: dialogue, thought, narration, aside, SFX, scene text/signage, heading/title, editorial note, credits, page number, watermark/logo text, ruby/furigana, unknown text. Punctuation-only text remains eligible.

Independent attributes: enclosure (speech/thought bubble, caption, uncontained, uncertain); script/language evidence (Japanese, Chinese, Korean, Latin, mixed, unknown, kept separate where possible); appearance (printed, handwritten, outlined, reversed, vertical, rotated, curved); relations (inside container, furigana parent, effect fragment, continuation); action (clean, keep, review) and mask quality.

These labels were proposed, not learned capabilities of the current models. They are not a prerequisite for reliable cleaning.

4. Corrections, qualifications and remaining choices

Corrections to carry forward

Q01 - “COO” is not a universal model. It is a dataset/research collection with multiple checkpoints. Detection polygons are not automatically lettering-accurate erase masks. See Document 04 and sources S22-S24.

Q02 - The example is Korean. Its Hangul lettering makes multilingual validation necessary. Japanese COO results cannot establish Korean/color-webtoon recall. This qualifies the earlier CTD + COO recommendation; it does not reject the experiment.

Q03 - The old path is not uniformly a filled rectangle. Source inspection shows existing pixel masks and a separate narrower model-ink path. The user-facing rectangular overlay is not proof that every backend path fills that rectangle. Preserve legacy behavior rather than replacing it with a newly invented rectangle algorithm. See Document 02.

Q04 - More padding alone is not the fix. First make the base mask match lettering; then grow that mask. More padding on a coarse blob increases artwork exposure.

Q05 - Mask quality and inpainting quality differ. A correct mask limits where the app edits. It does not guarantee exact reconstruction of artwork already hidden by opaque lettering.

Q06 - No performance claims were established here. No checkpoint inference, desktop execution, code changes or app benchmark was performed in this research session. Published scores are not measurements on the user's references.

Open choice register

ID	Unresolved choice	Recommended starting position
O01	Public mode names and default	“Current area (legacy)” and “Text-shaped + padding”; keep current default until explicit selection.
O02	Glyph-tight versus compact grouped footprint	Start from glyphs. Preview small optional grouping/closing; never force a convex hull.
O03	Padding default, range and scaling	Test 0, 1, 2, 3 and 5 source pixels; consider 2 px only as a trial default.
O04	Outlines, shadows and glows	Remove decoration belonging to lettering; retain uncertain broad glows for correction.
O05	Target scripts and art styles	Include Japanese and Korean at minimum in evaluation; determine other production priorities.
O06	Which SFX checkpoint ships	Decide after incremental recall, false positives, device cost and permission review.
O07	Whether a second segmenter is needed	Compare refined CTD with SAM-TS/Hi-SAM and the comic segmenter on actual masks.
O08	Per-device latency/memory budgets	Measure on the app's intended platforms before promising budgets.
O09	Outside-bubble permission and resume UX	Explicit all-text preset, visible scope, and a clear resumption policy; no silent global policy change.
O10	Licensing / checkpoint redistribution	Complete the exact artifact review in Document 08 before bundling.

Package relationship: Documents 02-08 expand these requirements and proposals. Where a later implementation choice differs, update this register rather than treating a previous suggestion as user approval.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 02

Research findings and code audit

What the references mean and what the current code actually supports

1. Reading the user's references



Desired: compact text-shaped footprint



Reference: rectangular overreach

User-provided conceptual references. The colored overlays are illustrative, not segmentation ground truth.

The desired reference uses a compact highlight around the upper-right sound effect and around the lettering in the lower balloon. The rectangular reference extends across surrounding art and the balloon boundary. The design goal is a text-shaped footprint with padding, not a slightly smaller rectangle.

The samples contain Korean lettering. They should be used as visual acceptance references, not fed into a benchmark as if the colored overlays were original text pixels. Obtain an unmarked source page and independently annotate removal masks for a meaningful model test.

Interpretation: disconnected components are allowed. The lower text can have a compact grouped footprint where a small margin connects nearby letters. No requirement has been established to erase all white space inside a word or paragraph envelope.

Important limit: these supplied images are illustrative. This report does not claim they show the actual masks emitted by the installed app, nor that either image was produced by a tested model.

2. Verified current architecture

The following findings describe publicly inspected source, not a reproduced desktop run.

Area	Source finding	Consequence for the new option
CTD adapter	Uses the bounding-box head and one-channel segmentation output; the two-channel line map is not used. Its box classes are weak language labels. [S02]	Add line-map recovery as a separately measured experiment. Never require a language class for the no-OCR policy.
Ogkalu support	Existing balloon helpers adopt uncovered text and seed masks, including an Otsu-style fallback. [S04]	Extend that candidate mechanism; do not reinstall an already-present capability. Audit fallback mask safety.
Mask representation	The Rust Mask stores binary membership inside rectangular storage bounds. [S07]	Arbitrary shapes are already representable. A rectangle can remain the storage extent without becoming the edit area.
Fitting	Fitted carries mask, ink and paper masks. Automatic growth and manual-authoritative paths differ. [S06]	Add an explicit geometry policy; do not overload a “manual” flag for automatic text-shaped regions.
Engine geometry	Model application grows the ink mask with an isolation margin; LaMa uses the narrower ink path. [S08, S09]	In the new mode, account for every effective margin before preview and commit.
Fill	Fill works with the fitted region and sampled paper. [S10]	Separate sampling extent from write support so fill cannot widen the approved footprint.
Compositor	Writes only through a patch's applied mask. [S11]	Reuse this invariant and ensure the applied mask is exactly the previewed support.

Policy findings

The documented automatic workflow gates bubble text to Japanese and holds outside-bubble text for review by default. The outside-bubble “Clean anyway” setting is separate from engine selection. [S05, S19]

The run's model readiness logic currently includes gate files. A real no-recognition policy therefore needs mode-aware capability checks and model initialization, not only an OCR checkbox. [S15, S18]

Do not conflate: not detected; detected but held; rejected by language policy; accepted but mask incomplete; and cleaned but inpainting poor. Each needs its own diagnostic outcome.

3. Why the overlay and the edit can disagree

Frontend finding. RegionLayer.svelte positions region buttons using bbox percentages and draws rectangular outlines. That is a geometric display of region bounds, not a rendering of the pixel mask. [S16]

API finding. ApiRegion includes bbox and mask metadata, while the examined ApiMask representation describes a layer/provenance record rather than transmitting the authoritative pixel footprint to this overlay. Add a versioned preview-mask reference or equivalent route. [S14]

Existing protection. The compositor already enforces changed pixels as a subset of applied patch masks. The new feature should strengthen the UI/backend agreement around that existing protection, not bypass the compositor. [S11]

Specific places to inspect while implementing

Verified path	Proposed responsibility
crates/cleaner-core/src/detect/mod.rs	Candidate and segmentation output contracts.
crates/cleaner-core/src/detect/boxes.rs	Keep bbox geometry separate from edit membership.
crates/cleaner-core/src/fit/mod.rs	Explicit legacy versus text-shaped geometry policy.
crates/cleaner-core/src/mask.rs	Deterministic round dilation and bounded mask operations.
crates/cleaner-core/src/engines/model.rs	New-mode write support, alpha and removal of hidden post-preview growth.
crates/cleaner-core/src/engines/lama.rs	Consume the prepared mask plan, not infer a new one.
crates/cleaner-core/src/engines/fill.rs	Read sampling context; write only through approved support.
crates/cleaner-core/src/composite.rs	Final enforcement and outside-mask regression tests.
src-tauri/src/run.rs	Policy-aware readiness, discovery and preview lifecycle.
src-tauri/src/library.rs	Versioned geometry metadata and mask preview transport.
src/lib/editor/RegionLayer.svelte	Draw actual text-shaped support and retain accessible selection.

These are change targets, not a completed patch. Additional denoise, manual-tool, export and state call sites must be traced in the implementation checkout.

4. Research conclusion: segmentation is the missing link

Baseline: use the CTD mask more effectively

Upstream CTD processes its line map, refines predicted masks against source-image evidence, and has an optional path for segmentation outside known blocks. The refinement combines candidate color/threshold components guided by the prediction. [S20, S21]

Recommendation: port or adapt the concept into a separately testable Rust refinement stage. Do not blindly copy all hole filling, dilation or thresholds: the new policy must control those operations and avoid double padding. Whole-component acceptance can leak into artwork when lettering touches an illustration.

Additional segmentation candidate: SAM-TS / Hi-SAM

This project explicitly supports pixel-level text-stroke segmentation as well as broader word, line and paragraph masks. The stroke output is the relevant candidate; a paragraph mask is not a safe substitute. Its reported evaluations are on scene-text datasets, not this Korean comic sample. [S27, S28]

Recommendation: compare a base-sized stroke checkpoint on difficult local crops. Its checkpoints require the corresponding SAM image encoder, and the documented workflow is a Python research stack. Do not assume the full model is a ready-made ONNX plugin. [S27]

Comic-specific alternative

The ogkalu YOLOv8m segmenter is described as trained on about 3,000 comic/webtoon images. The short model card does not establish cleanup-mask granularity or superiority to CTD. Inspect exported masks before choosing it. [S26]

What the research supports

Build the new mask option independently of a detector upgrade. Refined CTD is the first baseline; a specialized segmenter is a measurable optional enhancement. CTD + COO can address discovery, while segmentation and controlled compositing address rectangular overreach.

Research limitations: main/master may change; no immutable app commit or installed-build identity was verified. No models were executed, no synthetic score is presented, and no app source was modified. Source identifiers resolve in Document 08 and the per-document source links.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 03

Text-shaped cleaning algorithm

Mask construction, padding, context and an exact write-boundary contract

1. Two modes, one invariant for the new mode

Confirmed requirement: preserve the existing behavior and add a text-shaped option. The following algorithm is a proposed implementation.

Mode	Behavior
legacy	Execute the existing geometry and routing path unchanged. Do not reinterpret old records as filled rectangles.
text_shape	Build a lettering-derived footprint, add explicit padding, preview it, and constrain every committed edit to it.

The mode controls geometry, not the detector or engine. A CTD-only run can use text_shape. A CTD + COO run can use legacy. Both combinations should remain testable.

Separate the five geometries

Detection bounds: a rectangle/polygon used to localize a candidate, merge duplicates and request a crop.

Refinement context: surrounding original pixels inspected to distinguish lettering from background. It may exceed the candidate bounds.

Base lettering mask B: the best current estimate of text pixels, including text-owned outlines/shadows where established. It is immutable for a given revision.

Allowed write support W: B after the selected grouping and padding operations. This is the visible “box” the user asked to change into a shape.

Model context / hole: the crop and missing-pixel input used by the engine. They are implementation inputs, never permission to overwrite a larger region.

Required invariant

```
For every source pixel p outside W:  
  committed_output[p] = input_composite_below_this_layer[p]  
  
For the complete visible page:  
  changed_pixels_from_source is a subset of  
  the union of visible applied layer masks.
```

This is a sample-value guarantee, not a promise of identical compressed-file bytes. Metadata preservation is an independent export requirement. The current compositor already follows a mask-bounded design. [\[S11\]](#)

2. Build a trustworthy base mask

Proposed processing sequence

A1 - Discover. Run the selected detectors. Retain page-coordinate candidates and detector provenance; no text transcription is requested.

A2 - Establish context. Expand a local crop for refinement. For long strips, respect the verified source/decode window. Do not make the detection rectangle a hard glyph boundary when an outline extends beyond it.

A3 - Collect segmentation evidence. Use CTD's probability map. For COO-only or weak candidates, rerun segmentation on an enlarged local crop without requiring a second CTD box. Preserve confidence maps before thresholding where practical.

A4 - Refine against the source image. Evaluate foreground candidates using local color, polarity and component evidence guided by the text prediction. Cover dark text, white text and colored/outlined text. CTD's upstream implementation supplies a useful reference for this type of refinement. [S21]

A5 - Protect non-text. Flag components that extend into faces, speed lines, borders or broad same-colored artwork. A strong edge is evidence, not a universal stopping rule: letters themselves have strong edges. Avoid keeping an entire connected dark component just because one part intersects a text seed.

A6 - Keep small real components. Do not delete punctuation, detached Korean character parts, dots, furigana or isolated SFX fragments with a single global area threshold. Use scale-aware confidence and context.

A7 - Save B. Store a source-coordinate base mask and its provenance. If evidence is inadequate, return "mask needs correction," not a full rectangle.

Outlines, shadows and glows

The target is the lettering as rendered, not only its dark center. Include a text-owned outline or shadow in B when supported by the image. A broad translucent glow is ambiguous; present a preview and correction control rather than claiming perfect automatic separation.

Do not derive B from the cyan in the supplied references. Those overlays explain intent; they are not clean source pixels or trusted labels.

Failure ladder

Refined CTD -> optional second segmenter -> user mask correction / review. A fallback may reduce automation; it must not silently expand the edit to the full detected region.

3. Footprint, padding and topology

Default proposal: rounded expansion of the base

For source-coordinate pixel centers and a requested radius r :

$$W(r) = \{p \text{ in the valid source domain} : \text{distance}(p, B) \leq r\}$$

Use a Euclidean distance transform or a consistently defined circular structuring element. Record the convention, including how integer boundary pixels are treated. Dilation expands the foreground according to the structuring element; a square and a circle do not produce the same footprint. [S31]

Zero means zero. At $r = 0$, W equals B unless an explicitly selected grouping operation changes it. Avoid inherited minimum growth in `text_shape` mode.

Source pixels, not display pixels. Zooming, device-pixel ratio or detector resize must not change the applied padding. If a future relative/stroke-width option is added, display and persist the resolved source-pixel radius.

Rebuild from the base. Padding $2 \rightarrow 5 \rightarrow 2$ must reproduce the first 2-pixel mask exactly. Do not apply dilation to a previously dilated result or use an old applied mask as the new base.

Optional compact-group shape

The reference may be satisfied by disconnected padded glyphs or a gently connected text footprint. Start with the former. A small, previewed closing/gap-bridging parameter may connect nearby letters within the same text group; it must not connect distant SFX fragments or span a large artwork gap.

Do not use a bounding rectangle, minimum-area rectangle or global convex hull as the default erase shape. Preserve interior holes unless the user or a reviewed mask rule intentionally includes them. Closed character counters may contain real background art.

Parameter proposal, not settled defaults

Setting	Trial values	Rule
Edge padding	0, 1, 2, 3, 5 source px	2 px is a candidate default only.
Group-gap closing	Off initially	Separate from padding; show its extra area.
Feather	0-1 source px initially	Must remain inside W and not leave original glyphs visible.
Context expansion	Engine/refiner-dependent	Reading more context never authorizes more editing.

Do not add CTD's reference dilation, legacy first-step growth and a new user radius on top of each other. One mask plan must own the resolved geometry.

4. Engine and compositing contract

Build once; all engines consume the same plan

```
MaskPlan:
  base_mask_B
  write_support_W
  optional_blend_alpha_A
  refinement_context
  engine_context
  model_hole_H
  source_revision
  base_hash, support_hash, plan_revision
```

W is the maximum permitted footprint. A is zero outside W. At the confirmed text pixels, A must be one unless the user deliberately chose a partial-strength edit.

Recommended first version: $H = W$ and hard compositing ($A = 1$ in W). Add inward feathering only after measuring halos and residual text. A larger internal model hole can be a future experiment, but its output still cannot be committed outside W.

Inward feathering without leakage

A possible alpha rule is:

```
A(p) = 0          when p is outside W
A(p) = 1          when p is in B
A(p) = min(1, d_inside_W(p) / f)  otherwise, when f > 0
A(p) = 1          otherwise, when f = 0
```

Define pixel-center distances consistently. Test narrow strokes and thin padding. A feather wider than the available margin must not reduce opacity on text itself. Do not apply an unbounded blur and let nonzero alpha leak outside W.

Apply through a bounded patch

The engine may generate a rectangular output crop. Start the patch buffer as a copy of the source composite beneath the layer, blend only where A is nonzero, and store W as the authoritative applied mask. The final compositor must enforce W again.

For overlapping layers, rebuild a re-run against the intended lower-layer composite, excluding the layer being replaced. Keep the immutable source mask reference separate from the inpainting context reference. A padding reduction must restore previously covered pixels rather than bake them into a new source.

Existing isolation margins need explicit handling

The current model path has separate ink and isolation geometry. [S08] Simply setting Fitted.mask to a tight shape or passing manual=true is insufficient to prove that later stages will not expand it.

Add an explicit new-mode contract to model geometry, denoise, fill and any enabled per-region engine. When an engine cannot honor that contract, decline it visibly in this mode; never fall back to a wider hidden edit.

5. Preview identity, errors and implementation guidance

One authoritative mask revision

The backend prepares W and returns a mask reference plus source revision, mode, parameters and support hash. The canvas draws that mask. Apply requests identify the same revision. Reject a stale preview after a source, mask, padding, engine-geometry or grouping change.

Contours are useful for outlines, but a simplified contour is not the authority for writes. Use the raster footprint for fill/coverage and the contour only for presentation. A storage bbox or selection handle may remain rectangular without affecting W.

Validation before an engine runs

Check dimensions, crop transforms, finite parameters, clipping domain, nonempty support and source identity. Confirm B is contained in W. If protected artwork intersects B, report a conflict rather than silently shaving off lettering or erasing the protected mark.

Avoid treating detector confidence as mask correctness. Useful review diagnostics include mask area versus candidate area, how much refinement added, border/foreground ambiguity, unfilled outline evidence, and disagreement between mask sources. These are diagnostic features until calibrated.

Pseudocode contract

```
prepare(candidate, source, options):
    evidence = segment_with_context(candidate, source)
    B = refine_text_pixels(evidence, source, options.refinement)
    if B is empty or uncertain_beyond_policy:
        return NeedsCorrection(reason)
    B = apply_explicit_user_mask_corrections(B)
    W = derive_footprint_from_base(B, options)
    A = bounded_alpha(B, W, options.feather)
    assert B subset_of W
    assert support(A) subset_of W
    return immutable_plan(B, W, A, source.revision, options)

apply(plan, source_below, engine):
    require_matching_source_and_plan_revision(plan)
    generated_crop = engine.render(plan.context, plan.hole)
    patch = source_below.copy_crop(plan.W.bounds)
    blend_only_inside(patch, generated_crop, plan.A)
    commit_patch(patch, applied_mask=plan.W)
```

This is design pseudocode, not a compiled Rust patch. An implementation must integrate cancellation, color/depth handling, error propagation, source-window constraints and existing persistence APIs.

Completion criterion: preview, patch mask and exported applied mask refer to the same W; pixels outside the visible support remain unchanged for every engine supported by the new mode.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 04

Detection and model integration

CTD + COO, multilingual coverage and optional segmentation experiments

1. Recommended experiment, not a guaranteed winner

Working direction: CTD + a COO-trained DBNet++ detector, with no recognition stage. Retain ogkalu for its existing layout/candidate role. Feed all accepted candidates into the text-shaped mask workflow in Document 03.

The model components have different responsibilities:

Component	Role in the proposed system	Status
CTD	Baseline text discovery and mask evidence	Retain
ogkalu text/bubble detector	Enclosure and complementary text candidates	Retain
COO-trained DBNet++	Additional irregular SFX discovery	Benchmark first
Refined CTD mask	Initial lettering-mask baseline	Implement and evaluate
SAM-TS / Hi-SAM stroke output	Difficult-mask comparator	Optional experiment
PP-OCRv6 medium detector	Additional non-recognition text discovery	Reserve experiment
manga-ocr / TRBA	Reading text	Not required for this workflow

What COO's published evidence establishes

COO targets Japanese comic onomatopoeia and supplies text-region polygons and fragment links. The repository lists detection checkpoints and separate recognition/linking tasks. [\[S22\]](#)

COO checkpoint	Precision	Recall	Hmean
DBNet++	90.8%	60.9%	72.9%
MTSv3, detection-only evaluation	70.1%	66.0%	68.0%
ABCNetv2, detection-only evaluation	67.2%	65.1%	66.1%

These are repository-reported single-checkpoint results, not app measurements. DBNet++ has the highest Hmean among these rows, but not the highest recall. Different thresholds and combinations may change the relevant tradeoff. [\[S22\]](#)

No inference from these numbers establishes performance on Korean lettering. The user's color-webtoon example requires a separate Korean holdout. Do not ship a Japanese-only success claim as an “everything” result.

2. Candidate generation and fusion

Discovery passes

B1 - Preserve a full-page pass. It can see effects or layout extending beyond any one crop.

B2 - Add overlapping crops as a quality profile. Test the earlier 1024-1536 source-pixel, 15-25% overlap proposal against actual source dimensions and device budgets. Do not squeeze an entire very long webtoon into a small square and call that the only high-recall pass.

B3 - Evaluate unused CTD evidence. Implement line-map decoding and a conservative path for orphan segmentation components. Upstream demonstrates these mechanisms; their benefit in this app still needs measurement. [S20]

B4 - Run complementary discovery beyond existing boxes. A detector applied only within CTD-found regions cannot discover wholly missed text elsewhere on the page.

B5 - Map once to native coordinates. Store crop origin, letterbox/resize transform and source/decode-window identity. Reconcile overlap duplicates in page coordinates.

Fusion policy

Preserve credible candidates from either detector. Do not require ensemble agreement. Use per-model calibrated thresholds; raw scores from different architectures are not directly comparable.

Resolve duplicates using overlap plus text evidence and enclosure relationships. Do not suppress a text candidate because it lies inside a bubble candidate: the container and its text are different objects. Avoid merging two adjacent lines or unrelated effects solely because padded boxes intersect.

Preserve candidate lineage after merging, including model/checkpoint, raw score, crop and polygon. Store SFX as a proposed role only when justified; a generic “outside bubble” label does not establish sound-effect semantics.

Fragment handling without OCR

A visual effect can be one selectable group with several masks. Use spatial/appearance evidence or manual grouping; no transcription is required. Keep separate components across large gaps. Recognition and learned link prediction are not required to clean each fragment.

Failure outcome: a region discovered only by COO remains visible even when segmentation fails. Mark it “mask needed,” rather than dropping the detection or erasing its polygon automatically.

3. Segmentation candidates and their limits

DB polygons do not solve the mask requirement

DB's postprocessing converts a probability map into boxes/polygons and expands detected text regions. These are localization outputs, not a guarantee of exact glyph pixels. [S24]

Do not rasterize the entire COO polygon as the default new-mode erase area. Use it to select a crop, run a stroke segmenter and associate the resulting mask with the candidate. A rectangular crop is acceptable; a rectangular write mask is the behavior being avoided.

Comparator A: refined CTD

Start here because the app already has mask evidence. Measure source-image refinement and local segmentation reruns separately from detection additions. Keep CTD mask threshold, source refinement and user padding independently recorded.

Comparator B: SAM-TS / Hi-SAM

Use the **stroke-level** output, not a word, line or paragraph envelope. The official project provides automatic text-stroke inference and a sliding-window option. The demonstration distinguishes stroke segmentation from hierarchical detection. [S27, S36]

Choose a base-sized checkpoint for the first cost/quality experiment; consider the efficient variant only after output semantics and quality are checked. There is no verified out-of-box Korean-comic superiority, no app latency measurement, and no confirmed native export in this report.

Comparator C: ogkalu comic segmenter

Inspect mask granularity and actual coverage on outlined Korean and Japanese SFX. A model called "segmenter" may still use broader region annotations. The short card alone is insufficient evidence for replacing the current mask source. [S26]

Reserve discovery model: PP-OCRv6

The official medium detector has an ONNX release. A detector can be used without its recognizer. It remains an optional coverage comparison, not a required new stage or a promise of manga-specific SFX accuracy. [S30]

Promote only on incremental benefit

A model earns a production role when it adds meaningful recoveries or reduces correction time without unacceptable artwork exposure, latency or memory. Do not automatically load every researched model. Detailed semantic classification and VLM review remain deferred.

4. Prototype, export and deployment gates

Research prototype

Use an isolated Python environment for checkpoint evaluation and save plain candidate/mask outputs. Do not integrate a research stack into the desktop installer merely to prove the idea. Pin code, configuration and checkpoint hashes for reproducibility.

The DB implementation and COO's MTSv3/ABCNetv2 setups include research dependencies and compiled operations. [S23, S32, S33] Export success, supported operators and provider behavior must be established for the exact checkpoint, not inferred from a model family name.

Native adapter contract

The production adapter should return candidates with geometry, confidence, model identity and source transforms. A separate segmenter adapter should return per-pixel evidence with explicit semantics. Model input color order, normalization, padding, tensor names/shapes, output thresholds and coordinate conventions belong in the artifact manifest.

Validate Python-versus-export parity on fixed inputs before quality comparisons. Include CPU and the actual execution providers intended for each supported desktop platform. Compare outputs within declared numerical tolerances and inspect any changed candidate/mask decisions.

Optional model management

Use the app's model-manager pattern for download permission, checksums, on-demand residency, missing-model messaging and cancellation. A missing optional SFX detector must leave the baseline available. A no-recognition run must not be blocked by missing OCR or script-gate files.

Keep checkpoints and source-image rights separate. The COO DBNet++ checkpoint's redistribution status must be established explicitly; a code license or publicly downloadable weight is not a blanket clearance.

Required benchmark slices

Japanese monochrome; Korean color webtoon; ordinary bubble text; outlined/rotated/curved SFX; white and low-contrast text; tiny text/furigana; punctuation-only text; text crossing artwork; long-strip joins; mixed-script pages. Use unseen series where feasible.

Ship/no-ship rule: a specialist should materially improve the target corpus over the refined baseline. If COO fails the Korean slice, retain the plugin boundary and improve the dataset/model rather than asserting language independence.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 05

Backend and persistence integration

Additive architecture, versioned masks and unchanged legacy behavior

1. Treat this as a policy branch, not a global refactor

Confirmed: preserve the old option. **Proposed:** introduce an explicit `CleaningAreaMode` with legacy and `text_shape` variants. Missing fields in old records resolve to legacy.

Do not change existing global growth constants, morphology behavior or engine defaults to make the new mode work. Route `text_shape` through a dedicated mask-plan builder and leave the old geometry path intact. The current core already has pixel masks, patches and a bounded compositor that can be extended. [S07, S11, S12]

Independent settings

Axis	Proposed values	Compatibility rule
Cleaning area	legacy / text_shape	Old records and unchanged defaults use legacy.
Detection profile	existing / enhanced_sfx	Optional detector availability does not change mask semantics.
Text policy	legacy_japanese / all_detected_text	Only the latter bypasses recognition/script gating.
Outside bubbles	review / clean	Explicit scope decision; not inferred from an engine choice.
Engine	Existing supported choices	New mode adds no automatic cloud or heavy-engine escalation.

A convenience preset may choose `all_detected_text` plus outside-bubble cleaning after an explicit user action. It must show those choices and preserve a reproducible run snapshot.

Proposed extension points

New module: `text_mask` or equivalent, owning refinement, footprint derivation and immutable `MaskPlan` creation. The name is proposed; it does not currently identify an inspected file.

Existing modules: `detect/balloon` for candidates; `fit` for background statistics and routing; engines for bounded generation; `composite` for final enforcement; `project/patch` for artifacts and provenance. See the audited source map in Document 02.

Do not represent automatically generated text masks as hand-drawn solely to skip growth. Their source, behavior and provenance are different even when both have authoritative boundaries.

2. Data model and storage proposal

Keep object identity separate from geometry

Changing padding must not create a different region ID. Store an identity plus a revision. Bbox is the current storage/indexing extent, not the region's permanent identity or erase membership.

```
TextRegionPlan {
  region_id, source_id, source_revision,
  cleaning_area_mode, algorithm_version,
  detector_provenance[],
  candidate_geometry, enclosure, fragment_group_id?,
  base_mask_ref, base_mask_hash,
  applied_mask_ref, applied_mask_hash,
  alpha_ref?, alpha_hash?,
  padding_source_px, grouping_options, feather_source_px,
  refinement_context, engine_context,
  mask_quality_state, correction_revision,
  plan_revision
}
```

These are proposed fields, not the app's current schema. Integrate with existing sidecar buffers and provenance rather than duplicating the project store.

Keep three artifacts distinguishable

Base mask: lettering before the user's growth settings. It supports idempotent padding and correction.

Applied support: the exact allowed edit footprint that the preview displays. It is the compositor's mask.

Generated patch: engine pixels blended only within support, stored in the source's supported color/depth representation.

The current ApiRegion/ApiMask layer describes region geometry and provenance; it needs an additional reference to the actual prepared footprint for text-shaped previews. [\[S14\]](#)

Serialization and migration

Version the settings and mask-plan payload. Missing fields use legacy behavior; unknown future modes fail visibly rather than falling back to a rectangle. Existing patches can continue to render from stored buffers without loading any newly proposed model.

Converting an old region is explicit: recompute from the preserved source, preview the new plan and commit only after acceptance. Never relabel an old padded/applied mask as the new immutable base.

Persist enough to reproduce the chosen plan even if a model's defaults later change. Store detector and segmenter checkpoint hashes independently from the inpainting model hash.

3. Preview/apply protocol and state transitions

Proposed commands

prepare_text_mask(region, options, expected_source_revision) returns a prepared plan, mask preview reference, resolved parameters, quality state and plan hash.

apply_prepared_mask(plan_id, expected_plan_hash, engine) verifies source and plan identity, runs the engine with bounded geometry and commits a patch through W.

update_text_mask(plan_id, corrections, options) builds a new revision from the stored base or explicit edited base, not from the last applied support.

Command names are illustrative. Reuse the app's established command/event style and transport security rather than adding a parallel framework.

State vocabulary

```
candidate -> preparing -> ready_for_preview -> applying -> cleaned
               -> needs_mask_correction
candidate -> held_by_policy
applying  -> declined_by_engine / failed / cancelled
```

A preview artifact exists before a cleaned patch. Do not use “has patch” as the only way to represent a valid candidate mask. Keep skipped detections reviewable.

Race and failure behavior

Padding or source changes invalidate in-flight plans. Late model responses must not overwrite a newer revision. Cancellation before commit leaves the current visible layer unchanged. If preparing an optional specialist fails, explain the fallback detector profile without changing cleaning-area mode.

An engine cannot silently widen W to repair a quality failure. It can retry with another supported engine using the same plan, or offer a new, visibly larger plan for explicit acceptance.

Preview transport

Prefer a local mask artifact or tiled alpha texture addressed by region and revision rather than full-page binary arrays in JSON. Return a page-coordinate bbox plus mask dimensions/origin. Keep short-lived file/protocol access within existing local security boundaries.

The mask texture is authoritative; simplified contours are a display cache. The renderer must not resample a contour and then use that approximation to perform backend writes.

4. Engines, re-runs and compatibility

Engine integration checklist

Fill: sample sufficient clean surrounding paper, but paint only W. Sampling and write area are independent.

Denoise: it may inspect surrounding pixels, but any generated smoothing outside W must be discarded. Do not leave the current margin arithmetic implicitly active in text_shape.

LaMa: feed a suitable context crop and the planned hole. The existing narrower ink and isolation logic must be mode-aware; it cannot add an unpreviewed ring. [S08, S09]

Other enabled per-region engines: enforce the same final write support, honor existing permission and availability rules, or decline text_shape explicitly. This feature does not introduce an automatic cloud upload.

Quality gates: assess residual text and reconstruction without widening the mask as a hidden side effect. No OCR success/failure becomes an eligibility condition.

Re-running a layer

Remove the layer being replaced from the context composite. Regenerate its geometry from the stored base and corrections. Replacing a larger old mask with a smaller new one must reveal original/lower-layer pixels outside the new footprint.

Engine changes alone do not grow the mask. Undo restores both the patch and exact plan/settings revision. Redo restores the same artifacts, or a clearly recorded rerender when exact artifacts are unavailable.

Export and long strips

The displayed and exported page must use the same applied support. Lossless PNG/TIFF pixel comparisons can test outside-mask equality; do not use recompressed JPEG byte equality as the test.

Respect source/decode-window boundaries. Carry transforms through strip joins and split outputs back to the correct pages. A virtual gap or unverified join must never become an editable pixel region. Preserve current color mode, bit depth, profile and metadata handling; unsupported engine combinations remain explicit declines.

Rollback

Keep legacy path tests and feature flags. Disabling enhanced discovery must not disable text_shape. Disabling text_shape must not rewrite completed patches. Saved results remain viewable without the optional models that created them.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 06

Interface and user workflow

Make the actual cleaning footprint visible without removing the old controls

1. Mode controls and terminology

Proposed UI labels: “Current area (legacy)” and “Text-shaped + padding.” Final wording is open. The new mode must not be labeled as a guarantee of perfect detection or reconstruction.

Keep the current default initially. The user can choose text-shaped for a run or region. A run-level setting applies to newly prepared regions, not retroactively to every existing layer.

Core controls

Control	Behavior
Cleaning area	Switch between legacy and text-shaped.
Text-edge padding	Adjust the source-pixel margin around the base lettering mask.
Show affected area	Display the actual W footprint, not only bbox outlines.
Edit mask	Add/remove from the base or explicit correction layer.
Rebuild mask	Recompute from source evidence; preview before committing.
Engine	Existing cleanup engine selection; independent of area shape.

Detection profile and all-text/outside-bubble policy belong with run settings, not in the padding control. Recognition must not appear as a required step in the new workflow.

Visual contract

The filled/tinted overlay shows precisely which source pixels may be changed; an outline traces its edge. A bbox may still appear as a faint optional locator or selection handle, but it must be visually distinct from the affected-area overlay.

Keep mask opacity as a display preference only. It must not alter the cleaning strength or support. On low zoom, show a contour/locator for discoverability; at 100% allow an exact pixel view.

The current overlay uses bbox-positioned buttons and an accessible keyboard navigation model. Preserve those accessibility benefits while adding shape rendering and hit testing. [\[S16\]](#)

2. Selection, preview and correction

Region selection

A user can select a text group even if its footprint has several disconnected pieces. On pointer input, check the mask or a small display-only tolerance, not every pixel in a large bounding box. Keep keyboard and Layers-panel selection available for tiny masks.

A display-only hit tolerance must never expand the edit mask. If a selected region retains a rectangular keyboard focus ring, pair it with the true mask outline so the ring is not mistaken for the cleaning boundary.

Proposed interaction sequence

U1: choose cleaning-area mode and discovery/policy options.

U2: detect candidates and prepare text-shaped masks.

U3: inspect the actual affected-area overlay; show held/unsafe regions separately.

U4: adjust padding or correct the mask. Geometry updates immediately when derived from a cached base; a new segmentation pass shows a distinct preparing state.

U5: apply the selected engine through the displayed mask revision.

U6: compare with the original, correct residuals, or re-run without cumulative growth.

Corrections and tools

Support adding a missed outline or removing an artwork stroke. Manual mask edits are authoritative. Keep freehand/shape tools intact; merely selecting a text-shaped region must not reinterpret a user-drawn rectangle as automatic text.

Allow split/merge of selection groups without automatically filling the gaps between masks. Protect/exclude painting may be useful, but conflicts with known text must be visible rather than silently leaving letters behind.

Feedback and errors

Use specific messages: "Found; held by policy," "Text found; mask needs correction," "Optional SFX detector unavailable," "Preview changed; refresh before applying," or "Engine cannot use this source format." Do not collapse these into "No text found."

A mask with no trustworthy pixels cannot show a rectangle tinted as a ready-to-clean footprint. It remains a candidate marker with an explicit correction action.

3. Reference-specific acceptance and compatibility

Upper-right SFX

The preview should follow the Korean effect and its text-owned outline with the chosen margin. Avoid the large rectangular area over background figures and speed lines. Some background inside glyph counters or between strokes may remain untouched.

If the model cannot distinguish outline/glow from art, expose the uncertainty. This sample is a reason to test colored and outlined lettering, not proof that COO already handles it.

Lower bubble text

Keep the balloon outline outside the affected area. A compact padded footprint around the two text lines is acceptable; wiping the entire balloon or its surrounding rectangular crop is not the requested behavior.

A group footprint may connect nearby glyphs at the chosen radius. Do not connect the text to the balloon border through a thresholding artifact or broad automatic growth.

Old option

Switching back to legacy restores the old geometry workflow for subsequent operations. Existing layers are unchanged until explicitly rebuilt/reapplied. A chapter may contain legacy and text-shaped layers side by side.

The new mode should be usable with baseline models alone. The SFX detector is an independent optional enhancement. Neither a missing OCR file nor an uninstalled research segmenter should disable baseline text-shaped mask preparation when its required baseline models exist.

Proposed copy

Text-shaped + padding: "Clean within the detected text footprint and the margin you choose."

Padding: "Source-image pixels around the text. More padding includes more background."

Context: "Extra surrounding image used to estimate the background; it does not enlarge the affected area."

Uncertain mask: "Text was found, but its boundary needs correction before automatic cleaning."

These strings are draft copy. Implement through the app's translation catalog, preserve visible keyboard focus, and avoid changing the page position when a layer or toolbar control is used.

Done means: the user can distinguish detection bounds from the actual affected area, predict where an edit may occur, undo it, and continue using the old option unchanged.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 07

Validation and delivery plan

Acceptance tests, benchmark design and an incremental implementation backlog

1. Test the boundaries before judging the models

All tests below are proposed. No app tests or checkpoint benchmarks were run in this research session.

ID	Test	Required result
T01	Outside-mask writes, each supported engine	Zero changed source samples outside W for a single isolated layer.
T02	Page composition with overlapping layers	Changes are contained in the union of visible applied masks.
T03	Preview/apply identity	Applied support hash equals the accepted preview's support hash.
T04	Padding idempotence: 2 -> 5 -> 2	Final 2-pixel mask equals initial 2-pixel mask exactly.
T05	Zero padding	W equals the selected base footprint; no hidden minimum growth.
T06	Zoom / display scaling	Changing zoom or device-pixel ratio changes no source-coordinate mask bytes.
T07	Feathering	Alpha is zero outside W; confirmed text remains fully covered.
T08	Empty or uncertain base mask	Review/correction outcome; no filled-box fallback.
T09	Fragment and counter preservation	No automatic long bridge or filled artwork gap.
T10	Legacy regression	Same legacy path/results on pinned fixtures; no new default/model requirement.
T11	Missing OCR/script model in all-text policy	No-recognition workflow can start with its actual required models.
T12	Optional SFX model unavailable	Baseline remains usable and the selected fallback is explicit.
T13	Engine swap / retry	Mask geometry does not grow unless explicitly changed and previewed.
T14	Stale preview and cancellation	No old response overwrites a newer plan or cancelled edit.
T15	Padding shrink after applied edit	Old excess pixels are restored from the correct lower composite.

Test oracle

Compare decoded pixel samples, not screenshots, file bytes or perceptual similarity for boundary invariants. Treat image alpha, color depth and source transforms explicitly. For the user's marked references, test geometry intent only; actual mask quality requires an unmarked source.

2. Image quality and persistence cases

ID	Test family	Include
T16	Bubble borders	Text close to curved, jagged and thought-bubble outlines.
T17	Multilingual characters	Japanese, Korean and the other scripts actually intended for release.
T18	Tiny components	Furigana, punctuation, diacritics, detached Hangul parts.
T19	Text appearance	Black, white, colored, outlined, shadowed and low-contrast text.
T20	Difficult geometry	Curved, rotated, broken, very large and partially occluded SFX.
T21	Artwork contact	Letters touching hair, faces, screentone, speed lines and panel borders.
T22	Mask-source disagreement	CTD/other segmenter disagree; review is explicit and no blind broad union occurs.
T23	Round dilation definition	Known circles/corners at each tested radius; no square-growth surprise.
T24	Source edges / strip joins	Negative crop offsets, page edges, variable widths, verified and unverified joins.
T25	Color / depth / export	Supported grayscale/RGB/alpha, 8/16-bit and existing restricted-format behaviors.
T26	Persistence	Save/reopen without optional models; source and mask hashes still match.
T27	Mixed-mode chapters	Legacy and text-shaped patches coexist, export and undo correctly.
T28	Manual masks	Old brush/shape masks stay authoritative and unchanged.
T29	Selection/accessibility	Tiny/disconnected masks, Layers navigation, keyboard actions and click-to-apply.
T30	Resource pressure	Cancellation, out-of-memory, failed provider, model unload and long-strip residency.

Quality review rubric

Score residual lettering, missing outline/shadow removal, damaged non-text pixels, altered balloon/panel borders, reconstructed texture seams and invented marks separately. A mask can pass outside-boundary tests while still being too broad inside its boundary.

Do not claim that exact reconstruction under opaque SFX is objectively known on real pages without a clean reference. Use synthetic overlays on known clean backgrounds for paired reconstruction measurements and real pages for expert usability review.

3. Benchmark design and model promotion

Dataset proposal

Start with a few hundred diverse pages, then expand until rare classes have meaningful samples. Use unmarked, rights-cleared sources. Split by series/book where possible, and keep an untouched holdout outside model/threshold tuning.

Annotate candidate regions, actual lettering/removal pixels, text-owned outlines/shadows, protected artwork and selection groups separately. Keep ambiguous glow/occlusion cases marked as uncertain. Do not turn a text bounding polygon into glyph-mask ground truth.

Ablation sequence

A: current baseline. B: baseline + new mask workflow. C: B + overlapping discovery and CTD line/orphan recovery. D: C + COO specialist. E: best validated discovery + alternative segmenter. F: add PP detector only if needed.

Change one contribution at a time. Keep cleaning policy identical when measuring detector recall; otherwise a gate bypass can be mistaken for a detector improvement.

Metric	Meaning
Instance recall by slice	Fraction of intended text regions found, including SFX and bubble text separately.
False candidates per page	Extra review cost introduced by higher recall.
Complete-page recovery	Fraction of pages where every intended region is discovered.
Text-pixel recall	How much target lettering the base/applied mask covers.
Non-text exposure	Included non-target pixels, with intended padding reported separately.
Protected-art damage	Inclusion/change of annotated artwork or borders.
Correction effort	Human corrections and time to an acceptable page.
Runtime and peak memory	Per page/region on actual target hardware.

A useful exact invariant is $\text{leak_pixels} = \text{count}(\text{changed AND NOT W}) = 0$. For learned quality metrics, do not invent a universal pass threshold. Set targets after baseline measurement and product review; report uncertainty and sample counts by slice.

Promotion gate: improved SFX recovery or mask quality, no unacceptable regression on bubble text/Korean pages, tolerable review cost, device compatibility, and cleared deployment terms. Published Hmean on COO is not a substitute.

4. Implementation backlog and release sequence

Work item	Deliverable	Exit condition
P0 - Pin and diagnose	Record source commit, installed build, model hashes and outcomes on unmarked examples.	Detection, gate, mask and inpainting failures are separated.
P1 - Add geometry policy	legacy/text_shape mode and unit-tested bounded MaskPlan.	T01-T08 and T10 pass on synthetic fixtures.
P2 - Implement refinement	CTD-guided source-resolution base mask and idempotent padding.	Outlines/tiny components reviewed; no rectangular fallback.
P3 - Wire preview and persistence	Versioned mask artifact, region overlay, apply protocol, save/undo/reopen.	T03, T13-T15, T26-T29 pass.
P4 - No-recognition policy	Mode-aware model readiness and explicit all-text/outside-bubble preset.	No OCR/script dependency in that workflow; legacy untouched.
P5 - Improve discovery	Line-map/orphan/tile experiments and candidate fusion.	Incremental gains measured without quality-policy confounding.
P6 - Prototype COO	Exact checkpoint, comparison outputs and multilingual evaluation.	Useful complementary recall; no unverified shipping claim.
P7 - Native optional adapter	Export parity, provider checks, verified model artifacts.	Hardware, security and licensing gates complete.
P8 - Segmenter comparison	Refined CTD vs stroke-model/comic-segmenter experiments.	Add a model only if correction effort or damage improves.
P9 - Release and rollback	Feature-gated rollout, regression pack, user-facing docs.	Old workflow remains available; saved output survives disablement.

Parallel work: P6 research may run while P1-P4 are implemented. Production SFX integration should not bypass the safe-mask contract. P8 is optional, not a reason to block a useful baseline release.

Evidence to attach to the implementation PR

Pinned commit/model manifest; raw unmarked fixtures or authorized fixture references; before/base/padded/output masks; preview/applied hashes; per-slice benchmark counts; device/provider measurements; regression test output; licensing decision record; known failure examples.

Release claim discipline

Say “new optional text-shaped cleaning area” and describe measured coverage gains. Do not claim “all text in all languages,” “perfect SFX reconstruction” or “zero artwork damage” from a small benchmark. Zero writes outside the approved support is a testable boundary claim; correct content inside that support remains an image-quality question.

Source IDs link directly to primary references. Full evidence ledger: Document 08.

DOCUMENT 08

Sources, licensing and unresolved choices

Primary evidence ledger and the checks required before production bundling

1. Evidence and permission boundaries

Research was performed on 23 September 2026 using the primary sources listed below. App paths were inspected on public main; several upstream projects use main/master. An immutable app commit was not established. Re-pin all implementation references before modifying code or comparing releases.

The supplied images are user-provided conceptual references. They have not been treated as pixel-accurate labels, original model output or permission to publish a training dataset. No checkpoint was downloaded/executed for benchmarking and no repository changes were made.

License findings, not blanket legal clearance

Artifact	Verified statement / limitation	Required action
Current app / CTD	The project documents GPL-3.0 and CTD provenance. [S01]	Preserve applicable notices and source obligations; review modified distribution.
Ogkalu detector/segmenter	Model cards identify Apache-2.0. [S25, S26]	Verify exact checkpoint, runtime-code dependencies and packaging terms separately.
COO DBNet++	COO links a checkpoint and the DB codebase. A complete redistribution clearance for the exact artifact was not established here. [S22, S23]	Confirm weight license/provenance and permitted production use before bundling.
COO MTSv3	Repository points to CC BY-NC 4.0. [S32]	Do not assume an unrestricted commercial option.
COO ABCNetv2	Instructions distinguish academic use from commercial use and request contact for commercial use. [S33]	Obtain appropriate permission or exclude it from that deployment.
Hi-SAM	Repository license is Apache-2.0; model setup also needs SAM encoder weights. [S27, S29]	Check the complete checkpoint stack and data/weight terms, not only the repo license.
Manga109 / Manga109-s	Full dataset and commercial-use subset have different conditions, including redistribution restrictions. [S34]	Identify the actual source volumes/terms for any training or published artifacts.

Downloading a weight, converting it to ONNX, distributing it separately, or putting it behind an optional switch does not by itself settle its license. Review the exact planned distribution with appropriate licensing expertise where needed.

Artifact security proposal

Pin download URLs, file hashes, model version and expected tensor signatures. Evaluate unfamiliar Python checkpoints in an isolated environment. Do not package arbitrary executable checkpoint loaders into the desktop app. Track converter versions and the final exported-file hash.

2. Sources: application core

S01 Manga Cleaner: project overview and current model inventory

S02 Manga Cleaner: CTD adapter and output contracts

S03 Manga Cleaner: region geometry and grouping

S04 Manga Cleaner: balloon detection and adopted-text mask helpers

S05 Manga Cleaner: script and outside-bubble gate

S06 Manga Cleaner: mask fitting and separate ink mask

S07 Manga Cleaner: binary mask and dilation primitives

S08 Manga Cleaner: model applied mask, isolation and blending

S09 Manga Cleaner: LaMa engine

S10 Manga Cleaner: fill engine

S11 Manga Cleaner: compositor and outside-mask invariant

S12 Manga Cleaner: patches and provenance

S13 Manga Cleaner: project persistence

How these sources are used: to identify existing responsibilities and proposed change locations. Source comments can document intent or an earlier measurement without proving a behavior on the user's installed build. The report distinguishes those comments from measurements performed in this session.

3. Sources: interface, orchestration and upstream CTD

- S14 Manga Cleaner: API region and mask metadata
- S15 Manga Cleaner: run lifecycle and model preconditions
- S16 Manga Cleaner: rectangular region overlay and hit targets
- S17 Manga Cleaner: page layer stack
- S18 Manga Cleaner: toolbar and capability checks
- S19 Manga Cleaner: user-facing cleaning modes and defaults
- S20 CTD upstream: inference and optional undetected-mask refinement
- S21 CTD upstream: image-guided mask refinement
- S35 Manga Cleaner: growth, isolation and edit-margin constants

Implementation verification: trace all call sites in a pinned checkout, including denoise and manual operations. The listed paths are not a substitute for compiling, running regression tests or inspecting a real prepared mask.

Source provenance notes

The source ledger is supplied in machine-readable JSON in the package. Per-document source links point to the same primary references. No benchmark score in this package is represented as an experiment run against the user's app or images.

The PDF/Word copies preserve the user reference images only to explain the requested behavior. Do not publish those pages or add them to training material without the necessary rights.

4. Sources: detector and segmentation research

S22 COO: dataset, detection checkpoints and benchmark results

S23 DB/DBNet++: official implementation and installation

S24 DB: text-region polygon postprocessing

S25 ogkalu: text-and-bubble detector model card

S26 ogkalu: YOLOv8m comic text segmenter model card

S27 Hi-SAM: stroke segmentation, checkpoints and patch inference

S28 Hi-SAM paper: pixel, word, line and paragraph segmentation

S29 Hi-SAM: Apache-2.0 repository license

S30 PaddlePaddle: PP-OCrv6 medium ONNX detector

S31 OpenCV: dilation and structuring elements

S36 Hi-SAM: executable stroke segmentation demo

Interpretation rule: detection, stroke segmentation, word/paragraph region segmentation and inpainting are different tasks. A result or model name from one task does not establish accuracy on the others. Tests on scene text or Japanese manga do not establish Korean-color-webtoon performance.

5. Sources: permissions and final decision checkpoints

S32 COO MTSv3: installation and non-commercial license reference

S33 COO ABCNetv2: installation and commercial-use notice

S34 Manga109 / Manga109-s: official dataset terms

Choices still requiring product or engineering sign-off

O01-O04: mode names/default, compact-group behavior, resolved padding settings, and handling of outlines/shadows/glows. See the complete register in Document 01.

O05-O08: production language/style coverage, the exact SFX model, whether another segmenter is justified, and acceptable device costs. Decide from the benchmark, not model popularity.

O09-O10: explicit outside-bubble/resume policy and exact artifact permissions. Neither is implied by selecting a new mask shape.

Recommended next implementation baseline

Keep legacy unchanged. Add text_shape with refined CTD evidence and explicit source-pixel padding; render the backend's exact allowed support; enforce it at every engine/commit boundary; add an explicitly selected no-recognition all-text policy. Benchmark CTD + COO as the SFX extension and check Korean pages before promotion.

SAM-TS/Hi-SAM and the comic segmenter are mask-quality experiments. PP-OCRv6 is a reserve discovery experiment. Rich semantic classification, manga-ocr dependency, VLM review and large multi-model loading are not requirements for this feature.

What has not been decided or delivered

No final best-model claim, tuned threshold set, hardware budget, production license clearance, trained custom model, benchmark result on the sample, source-code patch, pull request, or app release is included. The delivered artifacts are research and implementation documents.

Definition of success: the new option finds more target lettering, produces an editable text-shaped footprint with a small explicit margin, cleans only within that footprint, and coexists with the original option.